ISLANDS AND CORAL REEFS / INDIA ISLANDS-GREAT NICOBAR

ISLAND-ORIGIN TREE → REEF-BUILDING ENGINE → REEF-TYPE CROSS-SECTION → DARWIN SEQUENCE → DALY QUALIFICATION → BLEACHING OUTCOME FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 ISLAND-ORIGIN TREE

ISLAND-ORIGIN TREE | CRUST OR SHELF AFFINITY | MAGMA BUILDS OCEANIC EDIFICE | REEF CARBONATE + WAVE-BUILT SEDIMENT | RIVER, CURRENT OR BARRIER ACCUMULATION

CONTINENTAL → crust or shelf affinity

VOLCANIC → magma builds oceanic edifice → CORAL → reef carbonate + wave-built sediment → DEPOSITIONAL → river, current or barrier accumulation

CLASSIFICATION
- CONTINENTAL → crust or shelf affinity
- VOLCANIC → magma builds oceanic edifice

DIAGNOSTIC BASIS
- CORAL → reef carbonate + wave-built sediment
- DEPOSITIONAL → river, current or barrier accumulation

PROCESS LINK
- CONTINENTAL → crust or shelf affinity
- VOLCANIC → magma builds oceanic edifice

EXAM TRAP
- CORAL → reef carbonate + wave-built sediment
- DEPOSITIONAL → river, current or barrier accumulation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CORAL → reef carbonate + wave-built sediment; DEPOSITIONAL → river, current or barrier accumulation.

01

STAGE 01 REEF-BUILDING ENGINE

REEF-BUILDING ENGINE | CALCIUM-CARBONATE SKELETON | PHOTOSYNTHETIC SYMBIOSIS | LIGHT + CLEAR WATER | REEF PRODUCTIVITY | SEDIMENT STRESS | BLEACHING OR DECLINE

POLYP → calcium-carbonate skeleton → ZOOXANTHELLAE → photosynthetic symbiosis → LIGHT + CLEAR WATER → reef productivity → HEAT / SEDIMENT STRESS → bleaching or decline

DEFINITION / COMPONENTS
- POLYP → calcium-carbonate skeleton
- ZOOXANTHELLAE → photosynthetic symbiosis

TRIGGER / MECHANISM
- LIGHT + CLEAR WATER → reef productivity
- HEAT / SEDIMENT STRESS → bleaching or decline

EFFECT / INTERVENTION
- POLYP → calcium-carbonate skeleton
- ZOOXANTHELLAE → photosynthetic symbiosis

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HEAT / SEDIMENT STRESS → bleaching or decline.

02

STAGE 02 REEF-TYPE CROSS-SECTION

REEF-TYPE CROSS-SECTION | SHORE CONTACT + NARROW LAGOON | OFFSHORE REEF + BROAD LAGOON | RING REEF AROUND LAGOON | LAND RELATION, LAGOON AND FOUNDATION

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
FRINGING → shore contact + narrow lagoon	ATOLL → ring reef around lagoon	FRINGING → shore contact + narrow lagoon	ATOLL → ring reef around lagoon
BARRIER → offshore reef + broad lagoon	TEST → land relation, lagoon and foundation	BARRIER → offshore reef + broad lagoon	TEST → land relation, lagoon and foundation

SYSTEM / DEFINITION
- FRINGING → shore contact + narrow lagoon
- BARRIER → offshore reef + broad lagoon

PROCESS / MECHANISM
- ATOLL → ring reef around lagoon
- TEST → land relation, lagoon and foundation

SPATIAL / INDIA PATTERN
- FRINGING → shore contact + narrow lagoon
- BARRIER → offshore reef + broad lagoon

HAZARD / LIMIT / RESPONSE
- ATOLL → ring reef around lagoon
- TEST → land relation, lagoon and foundation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ATOLL → ring reef around lagoon; TEST → land relation, lagoon and foundation.

03

STAGE 03 DARWIN SEQUENCE

DARWIN SEQUENCE | VOLCANIC ISLAND + FRINGING REEF | SLOW SUBSIDENCE + UPWARD CORAL GROWTH | BARRIER REEF + WIDER LAGOON | ISLAND SUBMERGED

VOLCANIC ISLAND + FRINGING REEF → SLOW SUBSIDENCE + UPWARD CORAL GROWTH → BARRIER REEF + WIDER LAGOON → ISLAND SUBMERGED → ATOLL

DEFINITION / COMPONENTS
- VOLCANIC ISLAND + FRINGING REEF
- SLOW SUBSIDENCE + UPWARD CORAL GROWTH

TRIGGER / MECHANISM
- BARRIER REEF + WIDER LAGOON
- ISLAND SUBMERGED → ATOLL

EFFECT / INTERVENTION
- VOLCANIC ISLAND + FRINGING REEF
- SLOW SUBSIDENCE + UPWARD CORAL GROWTH

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BARRIER REEF + WIDER LAGOON; ISLAND SUBMERGED → ATOLL.

04

STAGE 04 DALY QUALIFICATION

DALY QUALIFICATION | GLACIAL SEA-LEVEL FALL | EXPOSED PLATFORM | WAVE PLANATION | FOUNDATION SHAPING | POST-GLACIAL RISE | RENEWED REEF GROWTH | COMBINE SEA LEVEL, BIOLOGY AND TECTONICS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
GLACIAL SEA-LEVEL FALL → exposed platform	POST-GLACIAL RISE → renewed reef growth	GLACIAL SEA-LEVEL FALL → exposed platform
WAVE PLANATION → foundation shaping	VERDICT → combine sea level, biology and tectonics	WAVE PLANATION → foundation shaping

DEFINITION / COMPONENTS
- GLACIAL SEA-LEVEL FALL → exposed platform
- WAVE PLANATION → foundation shaping

TRIGGER / MECHANISM
- POST-GLACIAL RISE → renewed reef growth
- VERDICT → combine sea level, biology and tectonics

EFFECT / INTERVENTION
- GLACIAL SEA-LEVEL FALL → exposed platform
- WAVE PLANATION → foundation shaping

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → combine sea level, biology and tectonics.

05

STAGE 05 BLEACHING OUTCOME FORK

BLEACHING OUTCOME FORK | POLLUTION STRESS | SYMBIONT LOSS | PALE OR WHITE COLONY | STRESS EASES | POSSIBLE RECOVERY | STRESS PERSISTS | DISEASE, MORTALITY, EROSION

THERMAL / LIGHT / POLLUTION STRESS

SYMBIONT LOSS → pale or white colony → STRESS EASES → possible recovery → STRESS PERSISTS → disease, mortality, erosion

DEFINITION / COMPONENTS
- THERMAL / LIGHT / POLLUTION STRESS
- SYMBIONT LOSS → pale or white colony

TRIGGER / MECHANISM
- STRESS EASES → possible recovery
- STRESS PERSISTS → disease, mortality, erosion

EFFECT / INTERVENTION
- THERMAL / LIGHT / POLLUTION STRESS
- SYMBIONT LOSS → pale or white colony

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STRESS EASES → possible recovery; STRESS PERSISTS → disease, mortality, erosion.

06

STAGE 06 REEF-SERVICE CHAIN

REEF-SERVICE CHAIN | REEF COMPLEXITY | HABITAT + FISHERIES | REEF CREST | WAVE-ENERGY ATTENUATION | CARBONATE PRODUCTION | ISLAND SEDIMENT | LIVELIHOOD + EXPOSURE RISK

REEF COMPLEXITY → habitat + fisheries → REEF CREST → wave-energy attenuation → CARBONATE PRODUCTION → island sediment → DEGRADATION → livelihood + exposure risk

DEFINITION / COMPONENTS
- REEF COMPLEXITY → habitat + fisheries
- REEF CREST → wave-energy attenuation

TRIGGER / MECHANISM
- CARBONATE PRODUCTION → island sediment
- DEGRADATION → livelihood + exposure risk

EFFECT / INTERVENTION
- REEF COMPLEXITY → habitat + fisheries
- REEF CREST → wave-energy attenuation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CARBONATE PRODUCTION → island sediment; DEGRADATION → livelihood + exposure risk.

07

STAGE 07 INDIA ISLAND CONTRAST

INDIA ISLAND CONTRAST | TECTONIC ARC, HIGH RELIEF, SEISMIC SETTING | LOW CORAL ATOLLS AND REEFS | TEN DEGREE | NICOBAR DIVIDE | NINE DEGREE | MAIN LAKSHADWEEP DIVIDE

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
A&N → tectonic arc, high relief, seismic setting	TEN DEGREE → Andaman / Nicobar divide	A&N → tectonic arc, high relief, seismic setting	TEN DEGREE → Andaman / Nicobar divide
LAKSHADWEEP → low coral atolls and reefs	NINE DEGREE → Minicoy / main Lakshadweep divide	LAKSHADWEEP → low coral atolls and reefs	NINE DEGREE → Minicoy / main Lakshadweep divide

SYSTEM / DEFINITION
- A&N → tectonic arc, high relief, seismic setting
- LAKSHADWEEP → low coral atolls and reefs

PROCESS / MECHANISM
- TEN DEGREE → Andaman / Nicobar divide
- NINE DEGREE → Minicoy / main Lakshadweep divide

SPATIAL / INDIA PATTERN
- A&N → tectonic arc, high relief, seismic setting
- LAKSHADWEEP → low coral atolls and reefs

HAZARD / LIMIT / RESPONSE
- TEN DEGREE → Andaman / Nicobar divide
- NINE DEGREE → Minicoy / main Lakshadweep divide

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NINE DEGREE → Minicoy / main Lakshadweep divide.

08

STAGE 08 GREAT NICOBAR VALUE MAP

GREAT NICOBAR VALUE MAP | TECTONIC ISLAND | EARTHQUAKE-TSUNAMI EXPOSURE | BIOSPHERE RESERVE | CORE-BUFFER-TRANSITION LOGIC | REEFS + BEACHES + FORESTS | CONNECTED HABITATS | INDIGENOUS COMMUNITIES

TECTONIC ISLAND → earthquake-tsunami exposure

BIOSPHERE RESERVE → core-buffer-transition logic → REEFS + BEACHES + FORESTS → connected habitats → INDIGENOUS COMMUNITIES → rights and livelihood

SYSTEM / DEFINITION
- TECTONIC ISLAND → earthquake-tsunami exposure
- BIOSPHERE RESERVE → core-buffer-transition logic

PROCESS / MECHANISM
- REEFS + BEACHES + FORESTS → connected habitats
- INDIGENOUS COMMUNITIES → rights and livelihood

SPATIAL / INDIA PATTERN
- TECTONIC ISLAND → earthquake-tsunami exposure
- BIOSPHERE RESERVE → core-buffer-transition logic

HAZARD / LIMIT / RESPONSE
- REEFS + BEACHES + FORESTS → connected habitats
- INDIGENOUS COMMUNITIES → rights and livelihood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REEFS + BEACHES + FORESTS → connected habitats.

09

STAGE 09 PROJECT-STATUS FIREWALL

PROJECT-STATUS FIREWALL | PORT + AIRPORT + TOWNSHIP + POWER | CLEARANCE CONDITIONS | SAFEGUARDS TO IMPLEMENT | 12 FEB 2026 | FINAL FOREST APPROVAL NOT GRANTED | OFFICIAL CLAIM IS NOT MEASURED OUTCOME

PROPOSAL → port + airport + township + power

CLEARANCE CONDITIONS → safeguards to implement → 12 FEB 2026 → final forest approval not granted → RULE → official claim is not measured outcome

DEFINITION / COMPONENTS
- PROPOSAL → port + airport + township + power
- CLEARANCE CONDITIONS → safeguards to implement

TRIGGER / MECHANISM
- 12 FEB 2026 → final forest approval not granted
- RULE → official claim is not measured outcome

EFFECT / INTERVENTION
- PROPOSAL → port + airport + township + power
- CLEARANCE CONDITIONS → safeguards to implement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → official claim is not measured outcome.

10

STAGE 10 ISLAND APPRAISAL GATE

ISLAND APPRAISAL GATE | STRATEGIC AND ECONOMIC RATIONALE | HAZARD, FRESHWATER AND CARRYING CAPACITY | RIGHTS + CUMULATIVE EFFECTS | ALTERNATIVES, SAFEGUARDS, MONITORING

NEED → strategic and economic rationale

PLACE → hazard, freshwater and carrying capacity → PEOPLE / ECOLOGY → rights + cumulative effects → DECISION → alternatives, safeguards, monitoring

DEFINITION / COMPONENTS
- NEED → strategic and economic rationale
- PLACE → hazard, freshwater and carrying capacity

TRIGGER / MECHANISM
- PEOPLE / ECOLOGY → rights + cumulative effects
- DECISION → alternatives, safeguards, monitoring

EFFECT / INTERVENTION
- NEED → strategic and economic rationale
- PLACE → hazard, freshwater and carrying capacity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PEOPLE / ECOLOGY → rights + cumulative effects.

11

STAGE 11 ISLAND ANSWER SPINE

ISLAND ANSWER SPINE | ORIGIN AND REEF TYPE | ARC, ATOLL, CHANNEL AND LAGOON | HEAT, HAZARD, RIGHTS AND STATUS | ISLAND-SPECIFIC AND PRECAUTIONARY

CLASSIFY → origin and reef type → MAP → arc, atoll, channel and lagoon → ASSESS → heat, hazard, rights and status → VERDICT → island-specific and precautionary

DEFINITION / COMPONENTS
- CLASSIFY → origin and reef type
- MAP → arc, atoll, channel and lagoon

TRIGGER / MECHANISM
- ASSESS → heat, hazard, rights and status
- VERDICT → island-specific and precautionary

EFFECT / INTERVENTION
- CLASSIFY → origin and reef type
- MAP → arc, atoll, channel and lagoon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ASSESS → heat, hazard, rights and status; VERDICT → island-specific and precautionary.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | ARAKAN YOMA EXTENSION; TECTONIC/FOLD-BELT LINK. | NICOBAR ISLANDS BELONG TO THE TECTONIC ISLAND ARC AND | BARREN ISLAND | INDIA'S ONLY ACTIVE VOLCANO. | VOLCANIC ISLAND; BARREN ISLAND IS INDIA'S ONLY HISTORICALLY ACTIVE | ENTIRELY CORAL ATOLLS IN ARABIAN SEA. | A&N REGION LIES IN A MAJOR EARTHQUAKE ZONE.

OPTIONAL PROCESS DEPTH
- Andaman = Arakan Yoma extension; tectonic/fold-belt link.
- Nicobar islands belong to the tectonic island arc and are fringed by coral reefs; they are not simply coral atolls.
- Barren Island = India's only active volcano.
- Narcondam = volcanic island; Barren Island is India's only *historically* active volcano.

DATA / INSTITUTION LIMIT
- Lakshadweep = entirely coral atolls in Arabian Sea.
- A&N region lies in a major earthquake zone.
- Saddle Peak is in North Andaman and is the archipelago's highest point.
- NOT: Lakshadweep is volcanic like Barren Island → Lakshadweep is coral/atoll ; Barren Island is volcanic.

CONTEMPORARY PARALLEL
- NOT: Andaman Islands are coral like Lakshadweep → Andamans are linked to the Arakan Yoma fold chain .
- NOT: Narcondam is India's active volcano → Barren Island is active; Narcondam is dormant.
- NOT: Bay of Bengal and Arabian Sea islands have the same origin → Bay islands are tectonic/volcanic/coral mix; Lakshadweep is coral.
- NOT: Island geography is only physical → Indian islands also have strategic, ecological and disaster-risk dimensions.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels